



EAST WEST UNIVERSITY

***Exploratory, Statistical, And Structural Analysis
Scientific Retractions***

Course Title: Python Programming for AI and ML

Course Code: AIML503

Semester: Summer 2026

Section: 01

Submitted by

Name: Shairin Akter Hashi

ID: 2026-2-74-022

Date: 26/08/2026

Submitted to

Dr. Md Mostofa Kamal Rasel

Associate Professor

Department of Computer Science & Engineering

East West University

Table of Contents

Abstract.....	1
1. Introduction.....	1
1.1 Project objectives.....	1
2. Data and Analytical Methods.....	2
2.1 Dataset and analytical scope.....	2
2.2 Preprocessing strategy	2
2.3 Feature engineering.....	2
2.4 Enhanced analytical methods.....	3
3. Results.....	3
3.1 Data quality and missingness.....	3
3.2 Notice-type composition	4
3.3 Temporal pattern of recorded notices.....	4
3.4 Publisher concentration.....	5
3.5 Countries represented in author affiliations.....	5
3.6 Retraction reasons and multi-label structure.....	6
3.7 Publication-to-notice lag.....	7
3.8 Complexity and lag correlations.....	7
3.9 Reason co-occurrence.....	8
3.10 Identifier coverage and collaboration structure	8
4. Critical Analysis and Discussion	9
4.1 Why raw counts are not rates.....	9
4.2 Multi-label variables change interpretation	9
4.3 Partial years and right truncation.....	9
4.4 Large grouped notice events can distort trends.....	9
4.5 Statistical significance versus practical importance.....	10
4.6 Database architecture and DOI linkage	10
4.7 Taxonomy and coding changes	10
5. Methodological Limitations.....	10
6. Conclusion	10
References.....	11

Abstract

This project analyzes the Retraction Watch dataset using an end-to-end Python workflow designed around four core requirements: data preprocessing, exploratory data analysis (EDA), plotting, and critical interpretation. The uploaded dataset contained 71,641 rows and 21 source columns. After removing 241 completely empty rows and one fully empty export column, 71,400 valid records remained. Retractions formed 92.65% of the cleaned data, while expressions of concern, corrections, and reinstatements made up the remainder. Preprocessing treated placeholder identifiers and PubMed ID value 0 as missing rather than imputing fabricated values, parsed publication and notice dates, checked record-level duplication, and engineered timing and multi-value count features. EDA shows a highly uneven distribution of records across publishers, countries, subjects, article types, and retraction reasons. Publication-to-notice lag is strongly right-skewed, with a median of 1.36 years, while reason count, author count, and subject count have small positive Spearman associations with lag. The analysis also shows that reason tags frequently co-occur and that multi-country records represent about 16.2% of the cleaned dataset. The key critical finding is methodological: Retraction Watch counts cannot be interpreted as misconduct rates because publication-volume denominators are absent, the latest year is incomplete, and several fields are multi-label. The enhanced notebook therefore extends the standard EDA with effect-size, co-occurrence, concentration, linkage, and grouped-notice sensitivity analyses to support more defensible conclusions.

1. Introduction

Retractions and related update notices are part of the scholarly record. They can reflect errors, unreliable findings, authorship or policy issues, peer-review concerns, plagiarism, data problems, or other circumstances. Crossref makes the Retraction Watch dataset publicly available and updates it regularly. The dataset is especially useful for data-analysis practice because it combines categorical, date, identifier, and multi-label text fields in a large real-world table.

The purpose of this project is not to rank countries, publishers, or disciplines. Instead, it is to demonstrate a defensible analytical workflow: first improve data quality, then explore the data visually, then quantify selected relationships, and finally interpret the findings with appropriate limitations. This is important because a visually large count can be misleading when the underlying denominator, reporting practice, or database coverage differs across groups.

1.1 Project objectives

- Inspect the raw dataset structure, data types, missingness, and duplicate records.
- Perform conservative preprocessing without inventing missing identifiers or descriptive information.
- Create derived variables for retraction year, publication year, publication-to-notice lag, and multi-value field counts.
- Explore temporal, categorical, and structural patterns using appropriate visualizations.
- Examine reason co-occurrence, feature correlations, collaboration structure, and identifier coverage.
- Apply critical interpretation so that raw counts are not misrepresented as rates or causal effects.
- Provide an enhanced notebook that adds statistical effect sizes, enrichment, concentration, linkage, and sensitivity checks.

2. Data and Analytical Methods

2.1 Dataset and analytical scope

The executed notebook loaded a Retraction Watch CSV containing 71,641 rows and 21 source columns. The last rows revealed a block of completely blank export rows. After structural cleaning, 71,400 non-empty records remained. The latest notice date in the executed snapshot was 17 July 2026; therefore, 2026 is a partial year and should not be compared directly with complete calendar years.

Measure	Result
Raw CSV rows	71,641
Raw source columns	21
Completely blank rows removed	241
Valid cleaned records	71,400
Formal retractions	66,155 (92.65%)
Latest notice date	17 July 2026
Primary software	Python, pandas, NumPy, Matplotlib

2.2 Preprocessing strategy

The cleaning strategy was deliberately conservative. Completely empty rows and fully empty structural columns were removed. Text fields were stripped of surrounding whitespace, blank strings were converted to missing values, and date fields were parsed into datetime format. Identifier placeholders such as “Unavailable” were treated as missing. Retraction Watch uses PubMed ID value 0 when a PubMed identifier is unavailable, so 0 was converted to missing rather than retained as a genuine identifier. Exact duplicate rows and duplicate Record IDs were checked after cleaning.

Missing DOI, URL, and Notes values were not replaced with modes, averages, or arbitrary labels because these fields contain identifiers or descriptive information. Artificially filling them would create false data. Multi-value fields such as Country, Subject, Reason, ArticleType, and Author were handled as semicolon-separated lists, allowing both record-level and exploded analyses.

2.3 Feature engineering

The notebook created RetractionYear, RetractionMonth, OriginalYear, LagDays, LagYears, AuthorCount, CountryCount, SubjectCount, ReasonCount, ArticleTypeCount, DOI-coverage indicators, URL coverage, a multi-country collaboration flag, and broad time-period categories. These variables make it possible to compare not only raw categories but also case complexity and publication-to-notice timing.

2.4 Enhanced analytical methods

The enhanced notebook extends the standard EDA with additional methods intended to strengthen critical analysis. These include empirical cumulative distribution functions (ECDFs), Spearman rank correlation, Mann–Whitney U tests with rank-biserial effect sizes, Benjamini–Hochberg false-discovery-rate adjustment, recent-period reason prevalence comparisons, reason-pair support/Jaccard/lift, network visualization, subject–reason Fisher enrichment, publisher concentration metrics, grouped-notice threshold sensitivity, DOI-linked lifecycle checks, and batch-adjusted trend comparisons. These analyses are descriptive and associational rather than causal.

3. Results

3.1 Data quality and missingness

Variable	Effective missing
Notes	51,330 (71.89%)
Retraction PubMed ID	38,946 (54.55%)
Original Paper PubMed ID	37,997 (53.22%)
URLS	31,923 (44.71%)
Original Paper DOI	5,946 (8.33%)
Retraction DOI	4,667 (6.54%)
Retraction date	1 record
Original paper date	1 record

The missingness audit shows why data type and field meaning matter. PubMed ID fields initially appeared far more complete when 0 was treated as a number; after interpreting 0 as unavailable, more than half of these identifiers were effectively missing. DOI coverage was much stronger. Notes and URL fields were substantially incomplete, but this does not justify imputation because missing identifiers and missing descriptive notes cannot be reconstructed statistically.

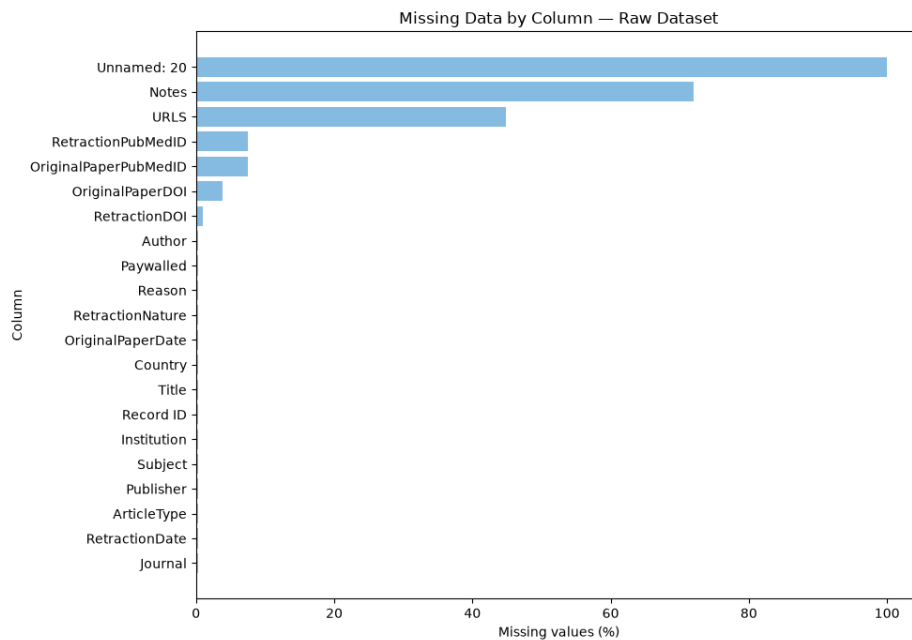


Figure 1. Missing values in the raw dataset before preprocessing.

3.2 Notice-type composition

The cleaned dataset is dominated by formal retractions. There are 66,155 retraction records (92.65%), 3,586 expressions of concern (5.02%), 1,499 corrections (2.10%), and 160 reinstatements (0.22%). This composition supports using retractions as the primary population for substantive statistical analysis. It also means that comparisons among all notice types should be cautious because Retraction Watch does not aim for comprehensive collection of corrections and expressions of concern.

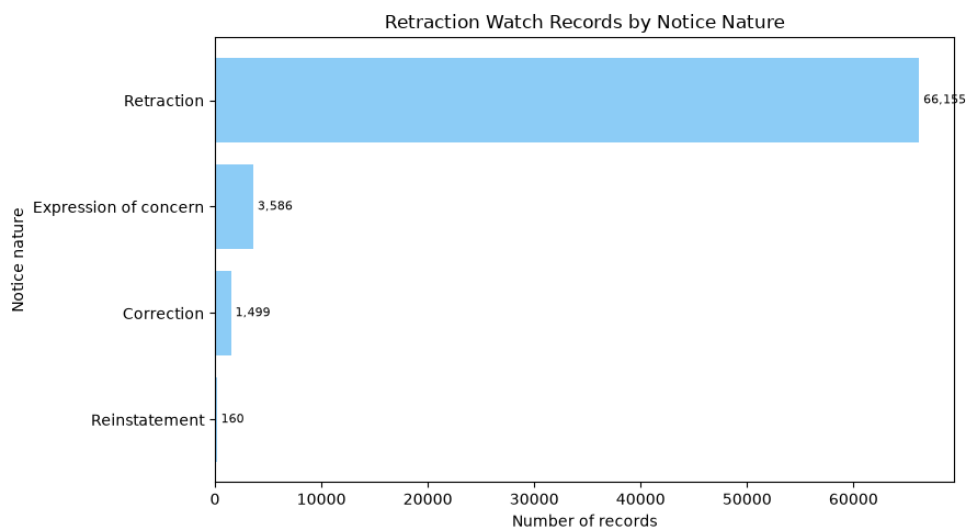


Figure 2. Distribution of Retraction Watch records by notice nature.

3.3 Temporal pattern of recorded notices

Annual counts increase strongly over the long run but do not form a smooth trend. The time series contains pronounced spikes around 2010–2011 and an especially large peak in 2023 at roughly 13.5 thousand records. Counts fall after the peak, while the 2026 value is visibly much lower because the dataset only covers part of the year. The correct interpretation is therefore not a simple continuous

increase or decline. Large publisher actions, retrospective additions, and grouped notices can generate sharp discontinuities in annual counts.

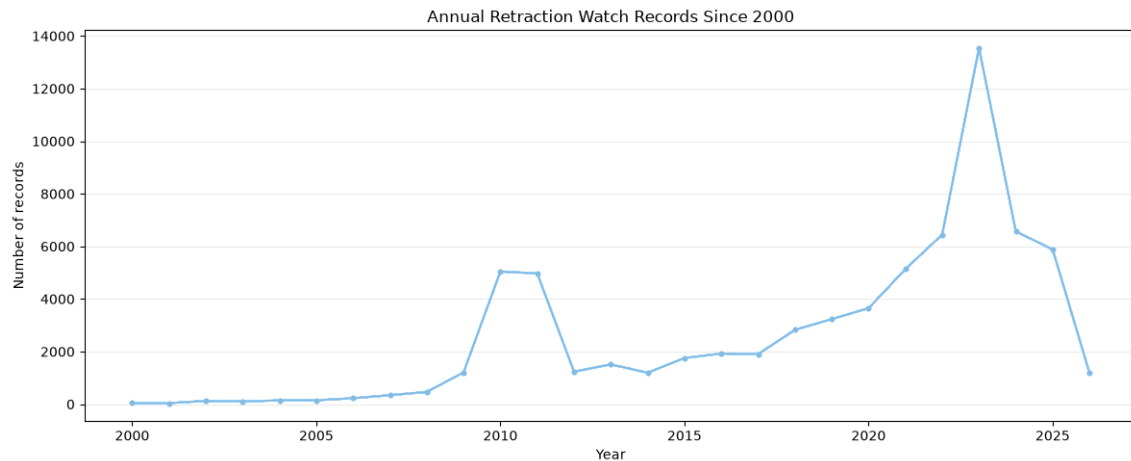


Figure 3. Annual Retraction Watch records since 2000. The 2026 value is a partial-year count.

3.4 Publisher concentration

The top publisher labels are highly concentrated. Hindawi has 11,524 records, followed by IEEE with 10,098, Elsevier with 7,195, Springer with 5,211, and Wiley with 4,289. Together, the top five publisher labels account for about 53.7% of cleaned records. This is a concentration result within the database, not a publication-normalized quality comparison. Publisher output volume, retrospective investigations, and large coordinated retraction actions can all influence these counts.

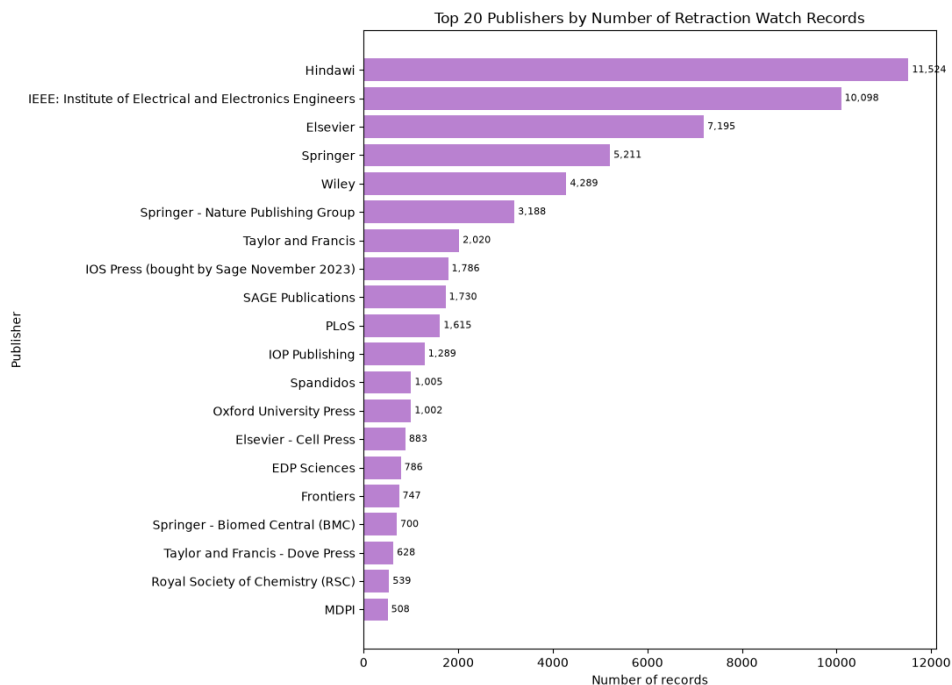


Figure 4. Top 20 publisher labels by number of Retraction Watch records.

3.5 Countries represented in author affiliations

China is the most frequently mentioned affiliation country with 37,774 mentions, followed by the United States (7,416) and India (6,282). Because Country is a multi-value field, one record may contribute to more than one country count. More importantly, these values are not national retraction

rates. A fair rate comparison would require total publication output by country, preferably adjusted for field, year, collaboration structure, and database coverage.

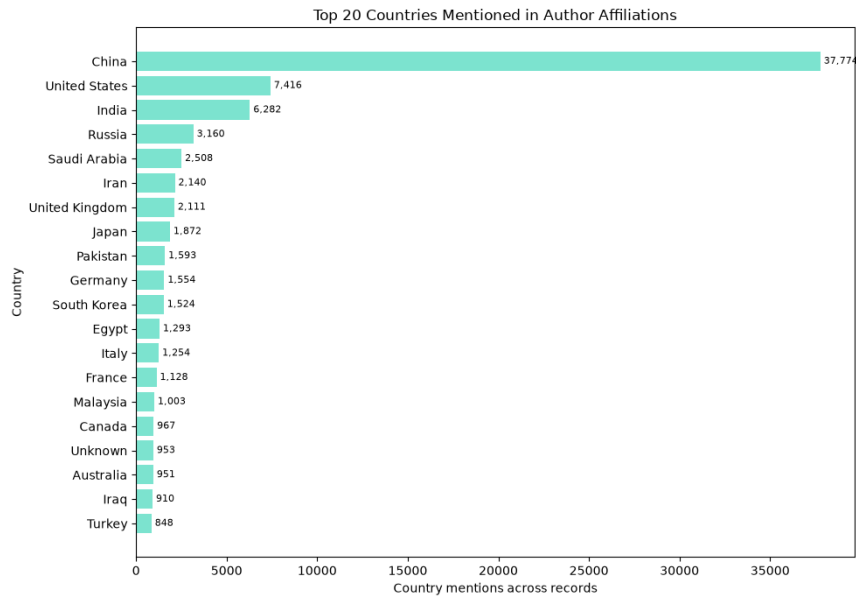


Figure 5. Top countries mentioned in author affiliations.

3.6 Retraction reasons and multi-label structure

The most common recorded reason is “Investigation by Journal/Publisher” with 31,500 mentions. It is followed by “Unreliable Results and/or Conclusions” (21,609), “Investigation by Third Party” (17,611), “Concerns/Issues about Data” (15,828), and referencing/attribution concerns (15,153). These counts do not sum to the number of records because Reason is a multi-label field. A single retraction may therefore reflect several procedural, evidential, or publication-related concerns at the same time.

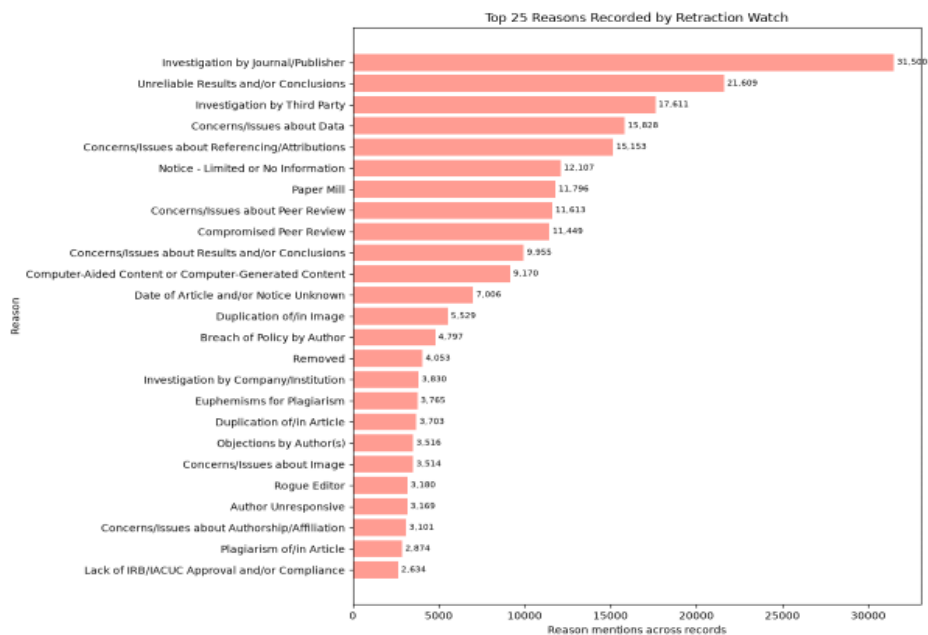


Figure 6. Top 25 reasons recorded in the dataset.

3.7 Publication-to-notice lag

The publication-to-notice lag is strongly right-skewed. The median is 1.36 years, while a long tail extends far beyond the median. This means the mean alone would be an unstable summary. Median, percentiles, boxplots, and ECDFs are more informative for this type of distribution. The long tail also has a substantive interpretation: some publications receive a formal update quickly, whereas others remain in the literature for many years before a notice appears.

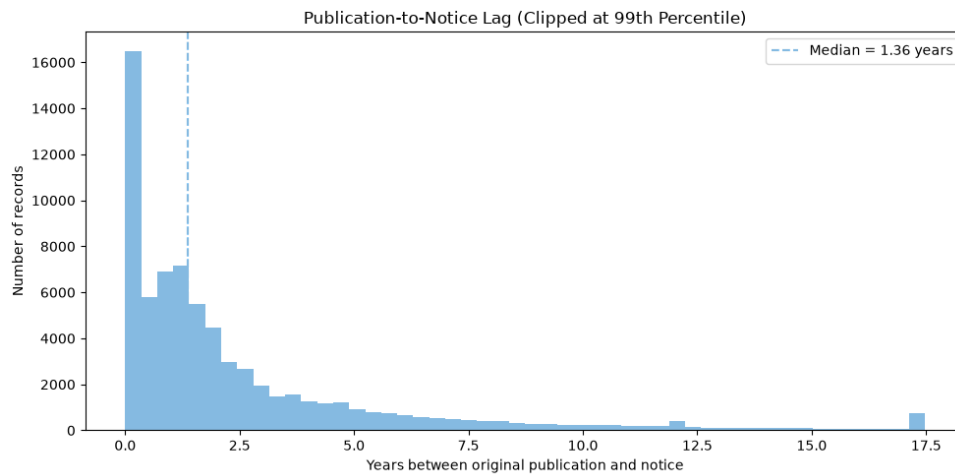


Figure 7. Publication-to-notice lag distribution, displayed through the 99th percentile.

3.8 Complexity and lag correlations

The engineered feature correlation matrix shows small positive associations between publication-to-notice lag and several indicators of record complexity. Spearman correlations with LagYears are approximately 0.24 for ReasonCount, 0.23 for AuthorCount, 0.20 for SubjectCount, and 0.10 for CountryCount. ArticleTypeCount is essentially unrelated to lag (about -0.01). These are modest associations, not strong predictive relationships. In a large dataset, even small correlations may be statistically significant, so practical magnitude is more important than the p-value alone.

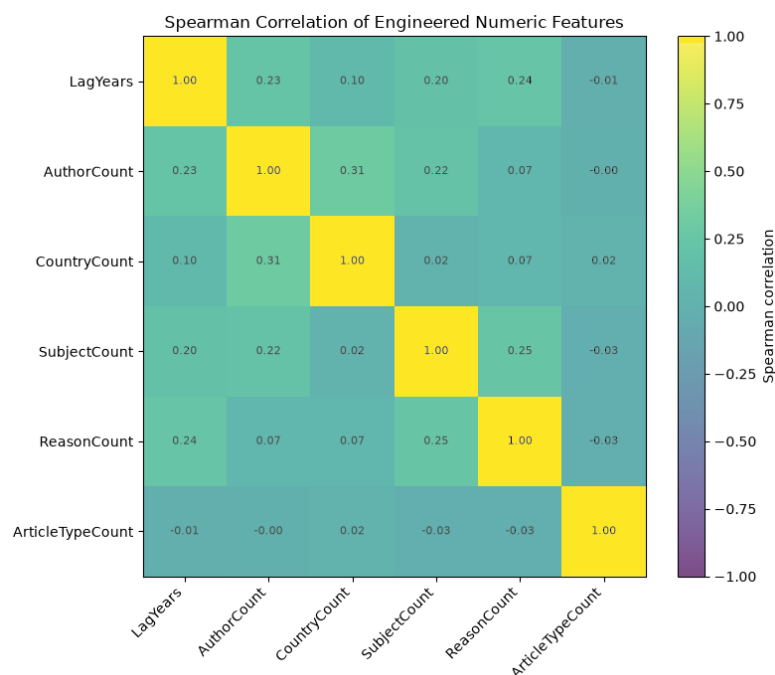


Figure 8. Spearman correlation matrix for engineered numeric features.

3.9 Reason co-occurrence

Retraction reasons frequently occur together. The most common pair in the executed notebook is “Investigation by Journal/Publisher” with “Unreliable Results and/or Conclusions” (18,298 co-occurrences). Other large pairs include journal/publisher investigation with third-party investigation (15,593), referencing/attribution concerns with journal/publisher investigation (13,911), and third-party investigation with unreliable results/conclusions (13,193). This demonstrates that reasons should not be treated as mutually exclusive categories. The enhanced notebook therefore computes association strength using Jaccard similarity and lift in addition to raw pair counts.



Figure 9. Top co-occurring pairs of recorded reasons.

3.10 Identifier coverage and collaboration structure

Identifier coverage is high for DOI fields but lower for Retraction Watch URLs. Retraction/update DOI availability is approximately 93.5%, original-paper DOI availability is about 91.7%, and URL availability is about 55.3%. These values indicate that DOI-based linkage is feasible for a large majority of records, although missing identifiers remain important for lifecycle and deduplication analyses.

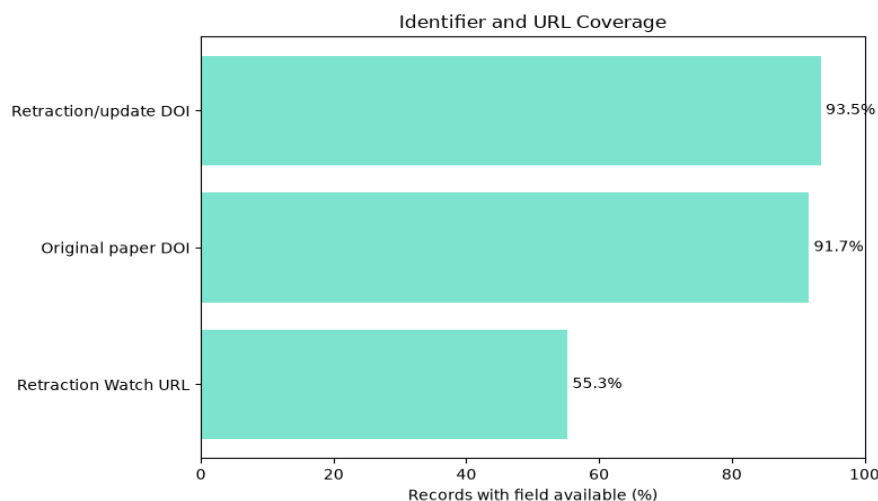


Figure 10. Coverage of DOI and URL fields in the cleaned data.

The collaboration flag identifies 11,567 records with affiliations from multiple countries and 59,833 records with a single or unknown country, so multi-country records make up about 16.2% of the cleaned dataset. This variable can be compared with timing and complexity, but it should not be interpreted as a causal effect of international collaboration.

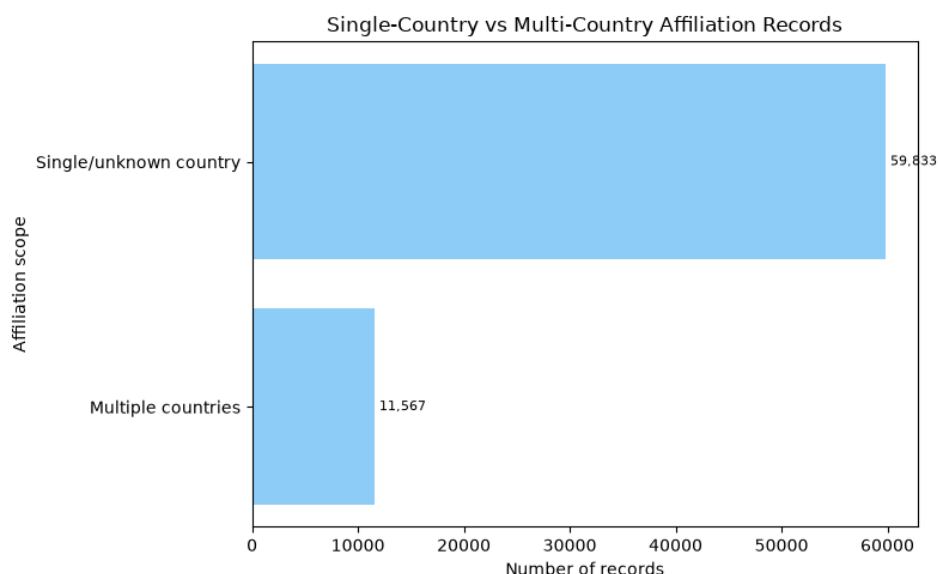


Figure 11. Single-country/unknown versus multi-country affiliation records.

4. Critical Analysis and Discussion

4.1 Why raw counts are not rates

The strongest methodological limitation is the absence of publication-volume denominators. For example, China has the largest affiliation count and Hindawi has the largest publisher count, but the dataset alone cannot tell us the proportion of publications that were retracted in either group. Larger publication volume, field composition, database coverage, investigation intensity, or major retrospective actions can produce large raw counts. The correct language is therefore “recorded counts in Retraction Watch,” not “retraction rate,” “misconduct rate,” or “quality ranking.”

4.2 Multi-label variables change interpretation

Country, Subject, Reason, ArticleType, Author, and Institution can contain several values in one record. Percentages based on these fields may overlap and should not be expected to sum to 100%. This also means that a reason is rarely an isolated explanatory variable. Co-occurrence analysis is more appropriate than a simple one-reason-per-record assumption.

4.3 Partial years and right truncation

The 2026 series is incomplete through 17 July, so the apparent decline at the end of the annual plot is partly a calendar artifact. Timing analysis has a related problem: recent publication cohorts have not yet had enough time to accumulate long-lag retractions. These issues make naive comparisons of the latest years potentially misleading.

4.4 Large grouped notice events can distort trends

The annual series contains abrupt peaks that are inconsistent with a smooth background process. One plausible structural explanation is that publishers sometimes issue many notices on the same date. The enhanced notebook therefore defines a transparent same-publisher/same-date grouped-notice proxy,

examines several minimum group-size thresholds, decomposes annual counts into grouped and other records, and compares recent reason shifts before and after excluding candidate grouped notices. This is a sensitivity analysis only; a grouped-notice proxy is not an official Retraction Watch event label and is not evidence of misconduct by itself.

4.5 Statistical significance versus practical importance

With more than 66,000 retractions, small differences can produce very small p-values. The enhanced analysis therefore reports rank-biserial effect sizes for Mann–Whitney comparisons, Spearman rho for monotonic associations, odds ratios for enrichment, and percentage-point changes for period comparisons. The interpretation should prioritize the magnitude and direction of an effect rather than using statistical significance as the only criterion.

4.6 Database architecture and DOI linkage

A Retraction Watch record is not necessarily identical to a unique scholarly paper. Repeated titles or DOIs can represent linked notices, multiple update stages, or group notices. Automatically deleting every repeated DOI would risk removing meaningful structure. The enhanced notebook therefore separates record-ID duplication from DOI/title multiplicity and includes DOI-linked notice lifecycle analysis.

4.7 Taxonomy and coding changes

Retraction reason terminology has changed over time. Retraction Watch documents additions and renamings in its database change log. A reason that becomes more common in recent years may therefore reflect a mixture of real-world change, database coding practice, vocabulary changes, and grouped investigations. Period comparisons are useful, but they should be treated as exploratory rather than causal evidence of changing research behavior.

5. Methodological Limitations

- No publication denominator is available for computing country, publisher, journal, article-type, or discipline retraction rates.
- Corrections and expressions of concern are not collected as comprehensively as retractions, so notice-type comparisons are not complete population comparisons.
- Several fields are multi-label and internally dependent; category counts and tests should not be treated as independent causal effects.
- The latest year is partial, and recent publication cohorts are right-truncated for long-lag events.
- Publisher, institution, and author strings are not fully entity-normalized, which can split or merge labels imperfectly.
- Reason vocabulary and coding conventions evolve, which can create apparent temporal shifts.
- Grouped-notice detection based on publisher and date is a structural proxy rather than an official event identifier.
- The project is observational and descriptive; it cannot establish causal relationships between record characteristics and retraction outcomes.

6. Conclusion

The project demonstrates a complete Python data-analysis workflow using a large, messy, real-world research-integrity dataset. The preprocessing stage removes structural noise while preserving meaningful missingness and multi-label information. EDA reveals strong concentration across publishers, countries, reasons, and time, while the lag distribution shows that formal updates occur on very different timescales. The correlation and co-occurrence results indicate that record complexity

and reason structure are meaningful analytical dimensions, but the observed relationships are generally descriptive rather than causal.

The central lesson is that critical analysis must accompany plotting. A large bar or a sharp temporal spike is not automatically evidence of poor research quality or increased misconduct. Publication volume, database coverage, grouped notices, incomplete years, changing taxonomy, and multi-label coding can all alter the story. For this reason, the enhanced notebook adds effect-size, linkage, enrichment, concentration, and sensitivity analyses so that conclusions are based on more than raw counts alone.

References

- [1] Crossref. Retraction Watch documentation. <https://www.crossref.org/documentation/retrieve-metadata/retraction-watch/>
- [2] Crossref. Retraction Watch Data repository. <https://gitlab.com/crossref/retraction-watch-data>